INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE CAMPUS

Degree: BCA, Stream: BCA

Term – I, Second Semester Examination, February - 2024

Subject Code: BCACC201

Subject Name: Computer Architecture

Full Marks: 30 **Duration: 1 Hour**

Part - A
Attempt 5 questions
Each question carries 2 marks (5*2)

1. Perform the arithmetic operations $(-42)_{10} + (-13)_{10}$ in binary using signed-2's complement representation for negative numbers. 2

   or

   Perform the subtraction $(100)_2 - (110000)_2$ with the following binary numbers using 1's complement. 2

2. What is high impedance state? 2

   or

   Write the significance of R1<- R2. 2

3. State the difference between logical shift and circular shift. 2

   or

   What is purpose of using Common Bus? 2

4. Define CPU. 2

   or

   Define SISD. 2

5. Define Speed-up in pipelining 2

   or

   Define efficiency in pipelining 2

Part - B
Attempt 2 questions
Each question carries 5 marks (2*5)

6. Explain the concept of One Address Instructions with the help of an example 5

   or

   Explain how a stack can be checked if it is full or empty 5

7. Draw the block diagram for the hardware for implementing the addition and subtraction operation and explain it. 5

or

Explain the concept of direct and indirect addressing with a diagram. 5

Part - C

Attempt 1 question

Each question carries 10 marks (1*10)

8. Draw the flowchart for signed add and subtract operations and analyse the different operations performed using it. 10

or

The 8-bit registers AR, BR, CR and DR initially have the following values: 10

AR = 11110010
BR = 11111111
CR = 10111001
DR = 11101010

Determine the 8-bit values in each register after the execution of the following sequence of microoperations.

AR<- AR + BR, CR<- CR ^ DR, BR <- BR + 1, AR<- AR-CR